

Whale-Watching Tourism vs. the Disturbance Behind It

1. The Issue

Whale watching is one of the fastest-growing forms of marine tourism worldwide. It allows people to encounter animals they may otherwise never see, but the closeness that makes the experience appealing also creates the problem: the vessels that make it possible can disturb whales through underwater noise, changes in behaviour, and physiological stress.

This report is one part of a larger group project examining the tension between whale watching tourism and the disturbance associated with it — a tension the International Whaling Commission has tried to manage since first issuing whale-watching principles in 1996 (International Whaling Commission, n.d.-a). My focus is the biological cost — the evidence for behavioural and physiological disturbance caused by vessels and their noise.

The broader social change question isn't whether whale watching is "good" or "bad" for whales. It's how the industry can be managed so people can keep interacting with whales without imposing unacceptable biological costs on the animals being watched.

2. Public Discourse and Media

The public conversation around whale-watching disturbance takes several forms, each reaching different audiences. How disturbance is communicated shapes what the public understands as the problem.

One form is viral incident coverage. In May 2026, footage circulated of a commercial whale-watching vessel sitting close to orcas near the Gulf Islands, BC. CTV/CP24 reported on the incident as Fisheries and Oceans Canada opened an assessment (CP24, 2026). The public can read the footage as visual evidence of harm, while the operator's response focused on compliance and the difficulty of judging distance through a telephoto lens. The regulator, meanwhile, has to balance whale protection against the economic and cultural benefits of coastal tourism.

Another form is regulatory discourse. Canada's 2026 expansion of protective distance rules for endangered Southern Resident killer whales requires vessels to stay 1,000 metres away (Fisheries and Oceans Canada, 2026). Unlike news coverage, this communication is procedural — rules, thresholds, instructions.

NGO advocacy communicates the issue differently again. Whale and Dolphin Conservation's "It's Rude to Intrude" campaign explains disturbance through relatable comparisons, such as being repeatedly woken up or having dinner interrupted, making disrupted feeding, resting, and nursing easier for a general audience to grasp (Whale and Dolphin Conservation, 2019).

Science-adjacent communication frames the issue differently still, as something manageable through technology. A May 2026 report on AI systems used to listen for endangered orcas presents acoustic disturbance as a problem that can be monitored and reduced (Mongabay, 2026).

Across these forms, the audience shapes what gets communicated: visual and emotional for viral coverage, procedural for regulators, relatable for NGOs, solutions-focused for technology. The most visible form of disturbance is not necessarily the most biologically important. A close approach can be filmed and shared immediately, while changes in respiration, stress hormones, or resting behaviour are largely invisible. This creates a communication gap between what people can see and what scientists measure.

3. Stakeholders

Biological disturbance from whale watching involves several stakeholders whose interests do not always align.

Whales and the marine ecosystem are unusual stakeholders: directly affected but unable to participate in public or policy discussion. Their interests — survival, undisturbed feeding, nursing, and breeding — must be represented by scientists, conservation organisations, and regulators.

Whale-watching operators and the tourism industry depend on successful whale encounters for financial viability. They need access to whales while also needing clear, workable regulations. This doesn't mean operators are automatically opposed to conservation, since the industry ultimately depends on healthy whale populations — but tourist expectations of close or reliable encounters can create pressure that contributes to disturbance.

Tourists and the general public are both consumers and a source of social pressure. They want close encounters for enjoyment, education, or connection with nature and also expect the experience to be ethical and environmentally responsible. Their expectations influence operator behaviour, while media coverage and conservation campaigns influence what they consider acceptable.

Marine scientists measure behavioural, physiological, and population-level responses. Their role is to identify measurable effects and determine what the evidence can actually support.

Government and regulatory bodies translate scientific evidence into enforceable rules while weighing tourism, local economies, and safety, which is a genuine challenge, since scientific evidence carries uncertainty while regulation requires decision-making about distances, speeds, vessel numbers, and licensing.

NGOs and conservation organisations advocate for whales and translate scientific findings into public facing arguments, increasing pressure for stronger protection, though their advocacy presents findings through a conservation lens.

These stakeholders overlap and are interconnected: research influences regulation, regulation affects operators, operators shape tourist experience, tourists drive demand, and NGOs and media shape public pressure on regulators. The International Whaling Commission frames whale-watching management the same way: as a multi-stakeholder process (International Whaling Commission, n.d.-b).

4. Relevant Evidence

The strongest evidence for biological disturbance comes from studies connecting vessel presence or noise with measurable physiological or behavioural change.

Rolland et al. (2012) used the sharp, unplanned drop in shipping traffic following 11 September 2001 as a natural experiment, comparing underwater noise with stress-related faecal hormone metabolites in North Atlantic right whales before and after the drop. Lower low-frequency noise was associated with lower stress hormone levels. This is useful evidence because vessel presence and noise normally change alongside other environmental factors, making them difficult to separate.

Lemos et al. (2022) examined a similar relationship in gray whales off the Oregon coast, comparing vessel counts and ambient noise with faecal glucocorticoid concentrations across three field seasons. Vessel traffic the day before sample collection was significantly associated with stress hormone levels, even after controlling for sex, age, and reproductive state.

Sprogis, Videsen and Madsen (2020) provide the strongest direct connection to whale watching. They played recorded vessel noise at different volumes to resting humpback mother-calf pairs off Western Australia. Louder playback reduced resting time by around 30%, doubled respiration rate, and increased swim speed by over a third. This controlled

experiment shows that vessel noise itself can produce measurable responses rather than just the presence of vessels.

Together, these studies cover three species, different regions, and both observational and experimental approaches. The evidence has limitations:

- Much of it is correlational, so causation isn't always established
- Sample sizes for large-whale studies are limited
- Responses may differ between species, populations, and contexts
- Short-term behavioural or physiological changes don't automatically demonstrate long-term effects on reproductive success or population survival

The evidence strongly supports the claim that vessels and their noise cause measurable disturbance in whales. Evidence linking these effects specifically to whale watching is strongest in controlled studies such as Sprogis et al. (2020), while broader vessel-noise research provides supporting context. It is still unclear to what extent short-term effects result in harm at the population level.

5. Discussion and Synthesis

Putting the discourse, stakeholders, and evidence together reveals a central tension: the evidence is complex and uncertain, while public and regulatory decisions often demand simple answers.

Viral footage can make a close approach look like the whole problem, while research shows acoustic disturbance and physiological stress can occur without being visually obvious. A minimum approach distance is a clear rule, but biological responses aren't only determined by distance; vessel noise, speed, duration of exposure, number of vessels, and whale activity all matter.

Stakeholders also carry different priorities: operators need clear, workable rules; tourists want meaningful encounters; scientists need to account for uncertainty; regulators need enforceable thresholds; conservation organisations generally favour precaution wherever biological harm is possible. Research on the Australian whale-watching industry found environmental managers prioritise species health and biological issues, while operators wanted clearer guidance on compliance, legislation, and rules (Dimmock et al., 2014).

Social change is unlikely to come only from producing more scientific evidence. Evidence also has to be communicated effectively and translated into something that stakeholders can actually understand and follow.

My position is that whale watching shouldn't necessarily be rejected, but should be managed around evidence-based limits on vessel proximity, noise, and interaction.

Management should also be adaptive, given the uncertainty around long-term population effects — rules monitored and revised as better evidence becomes available. This recognises the social and economic value of whale watching while acknowledging that the industry depends on healthy whale populations.

The social-change challenge becomes the industry's ability to adapt so that visiting whales does not threaten the biological conditions that make those experiences possible.

6. AI Use Acknowledgement

AI was used to help frame the references in the APA style, and cross check references and citations as well as for broken links.

chat link: <https://share.gemini.google/HXXzo3YEODoT>

7. References

CP24. (2026, May 12). *Video of B.C. whale watching boat beside orcas being assessed, DFO says*.

<https://www.cp24.com/news/canada/2026/05/12/video-of-bc-whale-watching-boat-beside-orcas-being-assessed-dfo-say/>

Dimmock, K., Hawkins, E., & Tiyce, M. (2014). Stakeholders, industry knowledge and adaptive management in the Australian whale-watching industry. *Journal of Sustainable Tourism*.

<https://www.tandfonline.com/doi/full/10.1080/09669582.2013.879311>

Fisheries and Oceans Canada. (2026). *Watching marine wildlife*.

<https://www.dfo-mpo.gc.ca/species-especes/mammals-mammiferes/watching-observation/index-eng.html>

International Whaling Commission. (n.d.-a). *Whale watching*.

<https://iwc.int/whalewatching>

International Whaling Commission. (n.d.-b). *Stakeholder engagement and adaptive management*. Whale Watching Handbook.

<https://wwhandbook.iwc.int/en/responsible-management/stakeholder-engagement-and-adaptive-management>

Lemos, L. S., Haxel, J. H., Olsen, A., Burnett, J. D., Smith, A., Chandler, T. E., Nieukirk, S. L., Larson, S. E., Hunt, K. E., & Torres, L. G. (2022). Effects of vessel traffic and ocean noise on gray whale stress hormones. *Scientific Reports*.

<https://www.nature.com/articles/s41598-022-14510-5>

Mongabay. (2026, May 22). *AI listens for endangered orcas to help reduce underwater noise exposure.*

<https://news.mongabay.com/short-article/2026/05/ai-listens-for-endangered-orcas-to-help-reduce-underwater-noise-exposure/>

Rolland, R. M., Parks, S. E., Hunt, K. E., Castellote, M., Corkeron, P. J., Nowacek, D. P., Wasser, S. K., & Kraus, S. D. (2012). Evidence that ship noise increases stress in right whales. *Proceedings of the Royal Society B.*

<https://royalsocietypublishing.org/rspb/article/279/1737/2363/74122/Evidence-that-ship-noise-increases-stress-in-right>

Sprogis, K. R., Videsen, S., & Madsen, P. T. (2020). Vessel noise levels drive behavioural responses of humpback whales with implications for whale-watching. *eLife.*

<https://elifesciences.org/articles/56760>

Whale and Dolphin Conservation. (2019). *It's Rude to Intrude.*

<https://wcl.org.uk/we-all-need-our-personal-space-so-do-whales-and-dolphins-its-rude-tointrude.asp>